```
z5477240@vx10:~$ dig www.amazon.com.au a

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> www.amazon.com.au a
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49459
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 1fefb8fccfac2d4c01000000685b689056d06c8be4b970f5 (good)
;; QUESTION SECTION:
;www.amazon.com.au.             IN      A

;; ANSWER SECTION:
www.amazon.com.au.      1107    IN      CNAME   tp.04f01a85e-frontier.amazon.com.au.
tp.04f01a85e-frontier.amazon.com.au. 35 IN CNAME www.amazon.com.au.edgekey.net.
www.amazon.com.au.edgekey.net. 16959 IN CNAME   e8474.dsca.akamaiedge.net.
e8474.dsca.akamaiedge.net. 10   IN      A       184.31.253.132

;; Query time: 0 msec
;; SERVER: 129.94.242.2#53(129.94.242.2) (UDP)
;; WHEN: Wed Jun 25 13:10:08 AEST 2025
;; MSG SIZE  rcvd: 221
```

Question 1: The IP address is 184.31.253.132

```
z5477240@vx10:~$ dig www.amazon.com.au cname

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> www.amazon.com.au cname
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 16553
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: ccf083c48cf7fd3501000000685b68afd4a0e0affbd98077 (good)
;; QUESTION SECTION:
;www.amazon.com.au.             IN      CNAME

;; ANSWER SECTION:
www.amazon.com.au.      1076    IN      CNAME   tp.04f01a85e-frontier.amazon.com.au.

;; Query time: 0 msec
;; SERVER: 129.94.242.2#53(129.94.242.2) (UDP)
;; WHEN: Wed Jun 25 13:10:39 AEST 2025
;; MSG SIZE  rcvd: 123
```

Question 2: The canonical name is tp.04f01a85e-frontier.amazon.com.au. Canonical names can be used to allow clients to make requests to larger servers since the IP address can change – the canonical name allows a client to make

requests to a main server without knowing the IP address. Since canonical names are hard to remember though, alias are easier to remember

Question 3: The cookie (DNS) is a cookie to ensure that the DNS request was not tampered with. The flag qr indicates that this DNS request is a query request, rd and ra indicate that the recursion (asking other DNS servers) is permitted. There is no aa flag which indicates that www.amazon.com.au is not part of the the DNS server we query. UDP indicates the packet type that the DNS makes. OPT PSEUDOSECTION refers to the relevant information regarding how the socket is protected.

Question 4: My local IP address of the local nameserver is 129.94.242.2.

Question 5:

```
z5477240@vx10:~/3331/lab3$ dig amazon.com.au ns

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> amazon.com.au ns
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31389
;; flags: qr rd ra; QUERY: 1, ANSWER: 8, AUTHORITY: 0, ADDITIONAL: 15

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 2d6480b9f37aed240100000068628c6680e0796ad9728302 (good)
;; QUESTION SECTION:
;amazon.com.au.                 IN      NS

;; ANSWER SECTION:
amazon.com.au.          3364    IN      NS      ns1.amzndns.org.
amazon.com.au.          3364    IN      NS      ns1.amzndns.com.
amazon.com.au.          3364    IN      NS      ns2.amzndns.com.
amazon.com.au.          3364    IN      NS      ns2.amzndns.org.
amazon.com.au.          3364    IN      NS      ns1.amzndns.co.uk.
amazon.com.au.          3364    IN      NS      ns1.amzndns.net.
amazon.com.au.          3364    IN      NS      ns2.amzndns.co.uk.
amazon.com.au.          3364    IN      NS      ns2.amzndns.net.

;; ADDITIONAL SECTION:
ns1.amzndns.co.uk.      540     IN      A       156.154.67.10
ns1.amzndns.com.        2009    IN      A       156.154.64.10
ns1.amzndns.net.        1102    IN      A       156.154.65.10
ns1.amzndns.org.        1506    IN      A       156.154.66.10
ns2.amzndns.co.uk.      258     IN      A       204.74.120.1
ns2.amzndns.com.        1018    IN      A       156.154.68.10
```

Name Server	IP
ns1.amzndns.org.	156.154.66.10
ns1.amzndns.com.	156.154.64.10
ns2.amzndns.com.	156.154.68.10
ns2.amzndns.org.	156.154.150.1
ns1.amzndns.co.uk.	156.154.67.10
ns1.amzndns.net.	156.154.65.10
ns2.amzndns.co.uk.	204.74.120.1
ns2.amzndns.net.	156.154.69.10

We can get these results by using the query NS.

Question 6:

```
z5477240@vx10:~$ dig -x 9.9.9.9

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> -x 9.9.9.9
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13074
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: e8c3c85ac07144e401000000685b6ca978e329a5cd65fd98 (good)
;; QUESTION SECTION:
;9.9.9.9.in-addr.arpa.          IN      PTR

;; ANSWER SECTION:
9.9.9.9.in-addr.arpa.   89942   IN      PTR     dns9.quad9.net.

;; Query time: 0 msec
;; SERVER: 129.94.242.2#53(129.94.242.2) (UDP)
;; WHEN: Wed Jun 25 13:27:37 AEST 2025
;; MSG SIZE  rcvd: 105
```

The DNS name associated with the IP address is dns9.quad9.net.. We can use the query dig -x 9.9.9.9 to get this information.

Question 7:

```
z5477240@vx10:~$ dig @129.94.242.2 yahoo.com

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @129.94.242.2 yahoo.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 18610
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 63928ad0492d1d3b01000000685b6cf9f11fc65c0ad51bc8 (good)
;; QUESTION SECTION:
;yahoo.com.                     IN      A

;; ANSWER SECTION:
yahoo.com.              199     IN      A       74.6.231.20
yahoo.com.              199     IN      A       74.6.143.26
yahoo.com.              199     IN      A       74.6.143.25
yahoo.com.              199     IN      A       98.137.11.164
yahoo.com.              199     IN      A       98.137.11.163
yahoo.com.              199     IN      A       74.6.231.21

;; Query time: 3 msec
;; SERVER: 129.94.242.2#53(129.94.242.2) (UDP)
;; WHEN: Wed Jun 25 13:28:57 AEST 2025
;; MSG SIZE  rcvd: 162
```

There is not an authoritative answer as there is no aa flag. This is because the server is not part of the local DNS, as such our local DNS does not have the ability to give an authoritative answer on behalf of another DNS.

```
z5477240@vx10:~$ dig tp.04f01a85e-frontier.amazon.com.au a

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> tp.04f01a85e-frontier.amazon.com.au a
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 61956
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 9713c0415c60076e01000000685b6dbcbbdf2d6bafbb0684 (good)
;; QUESTION SECTION:
;tp.04f01a85e-frontier.amazon.com.au. IN        A

;; ANSWER SECTION:
tp.04f01a85e-frontier.amazon.com.au. 34 IN CNAME www.amazon.com.au.edgekey.net.
www.amazon.com.au.edgekey.net. 15635 IN CNAME   e8474.dsca.akamaiedge.net.
e8474.dsca.akamaiedge.net. 10   IN      A       184.31.253.132

;; Query time: 0 msec
;; SERVER: 129.94.242.2#53(129.94.242.2) (UDP)
;; WHEN: Wed Jun 25 13:32:12 AEST 2025
;; MSG SIZE  rcvd: 190
```

Question 8:

```
z5477240@vx10:~$ dig @184.31.253.132 yahoo.com
;; communications error to 184.31.253.132#53: timed out
;; communications error to 184.31.253.132#53: timed out
;; communications error to 184.31.253.132#53: timed out

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @184.31.253.132 yahoo.com
; (1 server found)
;; global options: +cmd
;; no servers could be reached
```

The dig is unsuccessful since the Amazon name servers do not allow us to request information of Yahoo.

Question 9:

```
z5477240@vx10:~$ dig @68.180.131.16 yahoo.com

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @68.180.131.16 yahoo.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 64843
;; flags: qr aa rd; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1272
; COOKIE: 306417d95ef367e92650f59a685b6e33e02ab6d1ffc6643f (good)
;; QUESTION SECTION:
;yahoo.com.                     IN      A

;; ANSWER SECTION:
yahoo.com.              1800    IN      A       74.6.143.26
yahoo.com.              1800    IN      A       74.6.231.20
yahoo.com.              1800    IN      A       74.6.231.21
yahoo.com.              1800    IN      A       98.137.11.163
yahoo.com.              1800    IN      A       98.137.11.164
yahoo.com.              1800    IN      A       74.6.143.25

;; Query time: 155 msec
;; SERVER: 68.180.131.16#53(68.180.131.16) (UDP)
;; WHEN: Wed Jun 25 13:34:11 AEST 2025
;; MSG SIZE  rcvd: 162
```

As shown above an aa flag was sent in the response. We used the query: “dig @68.180.131.16 yahoo.com

Question 10:

Query to .

```
z5477240@vx10:~$ dig @198.41.0.4 lyre00.cse.unsw.edu.au ns

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @198.41.0.4 lyre00.cse.unsw.
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50666
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 5, ADDITIONAL: 11
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;lyre00.cse.unsw.edu.au.                IN      NS

;; AUTHORITY SECTION:
au.                     172800  IN      NS      t.au.
au.                     172800  IN      NS      r.au.
au.                     172800  IN      NS      a.au.
au.                     172800  IN      NS      s.au.
au.                     172800  IN      NS      q.au.

;; ADDITIONAL SECTION:
t.au.                   172800  IN      A       65.22.199.1
t.au.                   172800  IN      AAAA    2a01:8840:c1::1
r.au.                   172800  IN      A       65.22.197.1
r.au.                   172800  IN      AAAA    2a01:8840:bf::1
a.au.                   172800  IN      A       58.65.254.1
a.au.                   172800  IN      AAAA    2407:6e00:254::1
s.au.                   172800  IN      A       65.22.198.1
s.au.                   172800  IN      AAAA    2a01:8840:c0::1
q.au.                   172800  IN      A       65.22.196.1
q.au.                   172800  IN      AAAA    2a01:8840:be::1

;; Query time: 139 msec
;; SERVER: 198.41.0.4#53(198.41.0.4) (UDP)
;; WHEN: Wed Jun 25 13:47:49 AEST 2025
;; MSG SIZE  rcvd: 351
```

Query to .au.

```
z5477240@vx10:~$ dig @65.22.199.1 lyre00.cse.unsw.edu.au ns

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @65.22.199.1 lyre00.cse.unsw.edu.au ns
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 5442
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 3, ADDITIONAL: 7
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;lyre00.cse.unsw.edu.au.                IN      NS

;; AUTHORITY SECTION:
unsw.edu.au.            3600    IN      NS      ns1-ext.unsw.edu.au.
unsw.edu.au.            3600    IN      NS      ns2-ext.unsw.edu.au.
unsw.edu.au.            3600    IN      NS      ns3-ext.unsw.edu.au.

;; ADDITIONAL SECTION:
ns1-ext.unsw.edu.au.    3600    IN      A       54.79.80.189
ns1-ext.unsw.edu.au.    3600    IN      AAAA    2001:388:c:35::11
ns2-ext.unsw.edu.au.    3600    IN      A       13.236.238.52
ns2-ext.unsw.edu.au.    3600    IN      AAAA    2001:388:c:35::22
ns3-ext.unsw.edu.au.    3600    IN      A       54.66.99.146
ns3-ext.unsw.edu.au.    3600    IN      AAAA    2001:388:c:35::33

;; Query time: 3 msec
;; SERVER: 65.22.199.1#53(65.22.199.1) (UDP)
;; WHEN: Wed Jun 25 13:48:26 AEST 2025
;; MSG SIZE  rcvd: 249
```

Query to unsw.edu.au.

```
z5477240@vx10:~$ dig @54.79.80.189 lyre00.cse.unsw.edu.au ns

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @54.79.80.189 lyre00.cse.unsw.edu.au ns
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 21590
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 2, ADDITIONAL: 5
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
; COOKIE: 23353595b41b5abb0100000685b71a6fac77c9ea5051683 (good)
;; QUESTION SECTION:
;lyre00.cse.unsw.edu.au.                IN      NS

;; AUTHORITY SECTION:
cse.unsw.edu.au.        10800   IN      NS      beethoven.orchestra.cse.unsw.edu.au.
cse.unsw.edu.au.        10800   IN      NS      maestro.orchestra.cse.unsw.edu.au.

;; ADDITIONAL SECTION:
beethoven.orchestra.cse.unsw.edu.au. 10800 IN A 129.94.172.11
beethoven.orchestra.cse.unsw.edu.au. 10800 IN A 129.94.208.3
beethoven.orchestra.cse.unsw.edu.au. 10800 IN A 129.94.242.2
maestro.orchestra.cse.unsw.edu.au. 10800 IN A   129.94.242.33

;; Query time: 3 msec
;; SERVER: 54.79.80.189#53(54.79.80.189) (UDP)
;; WHEN: Wed Jun 25 13:48:54 AEST 2025
;; MSG SIZE  rcvd: 214
```

Querying cse.unsw.edu.au

```
z5477240@vx10:~$ dig @129.94.172.11 lyre00.cse.unsw.edu.au a

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> @129.94.172.11 lyre00.cse.unsw.edu.au a
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 15972
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 1edebefac2e5072e01000000685b71db941f514d5c0f19e6 (good)
;; QUESTION SECTION:
;lyre00.cse.unsw.edu.au.                IN      A

;; ANSWER SECTION:
lyre00.cse.unsw.EDU.AU. 3600    IN      A       129.94.210.20

;; Query time: 0 msec
;; SERVER: 129.94.172.11#53(129.94.172.11) (UDP)
;; WHEN: Wed Jun 25 13:49:47 AEST 2025
;; MSG SIZE  rcvd: 117
```

We had to use 3 ns queries and 1 a query to get an authoritative answer.

Question 11:

Yes, servers often have multiple IP addresses to deal with traffic or host multiple sites.